

Yanru Chen

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UCSD ECE Ph.D. candidate specializing in computer architecture and hardware/software co-design, with experience in heterogeneous memory and simulation for agentic AI, LLM inference, databases and graph analytics.

Technical focus: Python, C++, MATLAB, Verilog; architecture simulation, hardware/software co-design, processing-in-memory, heterogeneous memory and workload mapping.

EDUCATION

University of California San Diego Sep 2024 – Present
Ph.D. candidate in Electrical and Computer Engineering **Anticipated June 2027**
Advisor: Prof. Tajana Rosing; qualifying examination passed March 2026; GPA: 3.6/4.0.

Tsinghua University Aug 2021 – Jun 2024
M.S. in Electronic Information (Intelligent Manufacturing); GPA: 3.78/4.0.

Jilin University Aug 2017 – Jun 2021
B.E. in Electronic Science and Technology; College Honors; GPA: 3.53/4.0; rank: 6/108.

INDUSTRY EXPERIENCE

SK hynix Jun – Sep 2026
Emerging Memory AI Infrastructure Intern, AI Memory Solution / SOLAB San Jose, CA

GLAIM: agentic AI architecture and simulation *First author; submitted to ASPLOS 2027*

- Developed a trace-driven, event-driven simulator for agentic AI, integrating LLM prefill/decode, CPU tools, near-memory retrieval, persistent state and interconnect contention.
- Modeled hierarchical scheduling and compatible-request batching with dependency tracking, state ownership/versioning, copy-on-write sharing and data-readiness checks.
- Built a reproducible collection/replay workflow across five agentic and retrieval benchmarks; validated dependency ordering, resource reservations and data delivery in a portable simulator release.

ARCHITECTURE RESEARCH

Systems Energy Efficiency Lab, UC San Diego Feb 2025 – Present
Graduate Researcher; advisor: Prof. Tajana Rosing La Jolla, CA

CHIME — heterogeneous near-memory inference *DATE 2026; shared first author*

- Co-designed a chiplet architecture pairing M3D DRAM for attention with M3D RRAM for weights and FFN execution; mapped operators and KV-cache tiers to reduce cross-chiplet transfers.
- Simulation and RTL-based evaluation on FastVLM and MobileVLM reported up to 54× speedup and 246× energy efficiency over Jetson Orin NX.

HDDB — computational storage for SQL *DAC 2026; shared first author*

- Co-developed in-storage SQL processing that maps string and numeric predicates to hyperdimensional encodings, FeNAND similarity searches and filtering/aggregation processors.
- On TPC-DS fact-table filters, the modeled 1 TB design achieved 19.8× speedup over HeavyDB at SF=1; encoding tests preserved predicate/decode correctness with 10% randomly perturbed TLC cells.

ARCHITECTURE RESEARCH CONTINUED

RAPID-Graph — graph acceleration

DATE 2026; shared first author

- Co-developed recursive graph partitioning for exact all-pairs shortest paths, mapping Floyd–Warshall and min-plus kernels to in-place computation in PCM arrays.
- A cycle-accurate model reported $42.8\times$ speedup versus an estimated H100 baseline on 32,768-node graphs.

GEN-Graph — heterogeneous graph computing

First author; submitted to IEEE TCAD

- Co-designed PCM matrix and HBM traversal tiles for exact graph dynamic programming, matching dense APSP and irregular sequence-to-graph alignment to distinct dataflows.

PALUTE — lookup-table acceleration

ISLPED 2026; second author

- Contributed to an M3D DRAM accelerator that replaces inference arithmetic with in-memory lookups, combining near-memory table generation with tier-aware data placement and scheduling.

SELECTED PUBLICATIONS AND SUBMISSIONS

* Equal contribution.

ASPLOS 2027 <i>Submitted</i>	GLAIM: Glass-Panel Full-Stack System for Agentic AI Inference with Heterogeneous M3D Memory Y. Chen, P.-K. Hsu, S. Cho, et al.
DATE 2026	CHIME: Chiplet-based Heterogeneous Near-Memory Acceleration for Edge Multimodal LLM Inference Y. Chen*, R. Tian*, Y. Pan, Z. Li, W. Xu, and T. Rosing.
DAC 2026	HDDB: Efficient In-Storage SQL Database Search Using Hyperdimensional Computing on Ferroelectric NAND Flash Q. Zhao*, Y. Chen*, R. Tian, et al.
DATE 2026	RAPID-Graph: Recursive All-Pairs Shortest Paths Using Processing-in-Memory for Dynamic Programming on Graphs Y. Chen*, Z. Li*, K. Fan, R. Tian, J. Hsu, W. Xu, M. Zhou, and T. Rosing.
ISLPED 2026	PALUTE: Processing-In-Memory Acceleration via Lookup Table for Edge LLM Inference R. Tian, Y. Chen, W. Xu, and T. Rosing.

EARLIER RESEARCH AND ENGINEERING

UC San Diego, Flexible and Printed Electronics Lab (2024–2025): Memristor fabrication and multimodal sensing; MEMS/NEMS and biomedical signal processing.

Tsinghua University (2021–2024); Hanit, Assistant Engineer (2023): Wearable acetone sensors, laser-induced graphene strain sensors, microfabrication and LiG–SiC nanosheets.

Western University (2020); Vivolight, Algorithm Intern (2021): CT perfusion and fractal analysis, 3D perfusion reconstruction, and OCT/NIRS research for lipid imaging.

SELECTED HONORS

UCSD CSE Best Doctoral Research Award	2026
DAC Young Fellow Award	2026
Jacobs Fellowship, University of California San Diego	2024
Second Best Oral Presentation, Workshop on Learning and Information Theory	2024